EE2001 Electrical Circuits- I:

Natural and Step Responses of RLC Circuits

09 January 2024

Outline



1. Introduction to the Natural Response of a Parallel RLC
2. The Forms of the Natural Response of a Parallel RLC Circu
3. The Step Response of a Parallel RLC Circuit
4. The Natural and Step Response of a Series RLC Circuit

Introduction to the Natural Response of a Parallel RLC

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The first step in finding the natural response of the circuit shown in Fig. 8.1 is to derive the differential equation that the voltage v must satisfy. We choose to find the voltage first, because it is the same for each component. After that, a branch current can be found by using the current-voltage relationship for the branch component. We easily obtain the differential equation for the voltage by summing the currents away from the top node, where each current is expressed as a function of the unknown voltage v:

$$\frac{v}{R} + \frac{1}{L}\int_0^t v\,d\tau + I_0 + C\frac{dv}{dt} = 0. \tag{8.1}$$

We eliminate the integral in Eq. 8.1 by differentiating once with respect to t, and, because I_0 is a constant, we get

$$\frac{1}{R}\frac{dv}{dt} + \frac{v}{L} + C\frac{d^2v}{dt^2} = 0. \tag{8.2}$$

We now divide through Eq. 8.2 by the capacitance C and arrange the derivatives in descending order:

$$\frac{d^2v}{dt^2} + \frac{1}{RC}\frac{dv}{dt} + \frac{v}{LC} = 0. \tag{8.3}$$

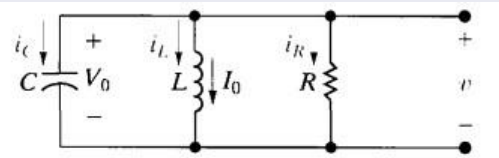


Figure 8.1 ▲ A circuit used to illustrate the natural response of a parallel *RLC* circuit.

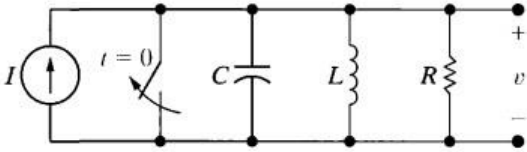


Figure 8.2 ▲ A circuit used to illustrate the step response of a parallel *RLC* circuit.

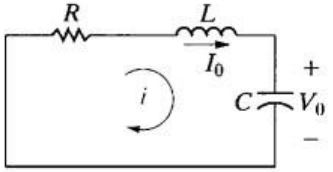


Figure 8.3 ▲ A circuit used to illustrate the natural response of a series *RLC* circuit.

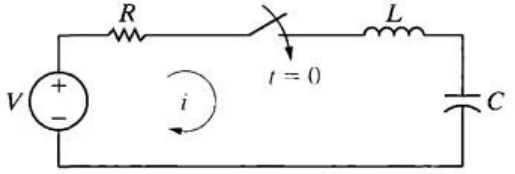


Figure 8.4 ▲ A circuit used to illustrate the step response of a series *RLC* circuit.

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$$\frac{d^2v}{dt^2} + \frac{1}{RC}\frac{dv}{dt} + \frac{v}{LC} = 0. \tag{8.3}$$

Comparing Eq. 8.3 with the differential equations derived in Chapter 7 reveals that they differ by the presence of the term involving the second derivative. Equation 8.3 is an ordinary, second-order differential equation with constant coefficients. Circuits in this chapter contain both inductors and capacitors, so the differential equation describing these circuits is of the second order. Therefore, we sometimes call such circuits **second-order circuits**.

The General Solution of the Second-Order Differential Equation

We can't solve Eq. 8.3 by separating the variables and integrating as we were able to do with the first-order equations in Chapter 7. The classical approach to solving Eq. 8.3 is to assume that the solution is of exponential form, that is, to assume that the voltage is of the form

$$v = Ae^{st}, \tag{8.4}$$

where A and s are unknown constants.

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$$v = Ae^{st}, \tag{8.4}$$

If Eq. 8.4 is a solution of Eq. 8.3, it must satisfy Eq. 8.3 for all values of t. Substituting Eq. 8.4 into Eq. 8.3 generates the expression

$$As^2e^{st} + \frac{As}{RC}e^{st} + \frac{Ae^{st}}{LC} = 0,$$

or

$$Ae^{st}\left(s^2 + \frac{s}{RC} + \frac{1}{LC}\right) = 0, \tag{8.5}$$

which can be satisfied for all values of t only if A is zero or the parenthetical term is zero, because $e^{st} \neq 0$ for any finite values of st. We cannot use $A = 0$ as a general solution because to do so implies that the voltage is zero for all time—a physical impossibility if energy is stored in either the inductor or capacitor. Therefore, in order for Eq. 8.4 to be a solution of Eq. 8.3, the parenthetical term in Eq. 8.5 must be zero, or

$$s^2 + \frac{s}{RC} + \frac{1}{LC} = 0. \tag{8.6}$$

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$$s^2 + \frac{s}{RC} + \frac{1}{LC} = 0. \tag{8.6}$$

Equation 8.6 is called the **characteristic equation** of the differential equation because the roots of this quadratic equation determine the mathematical character of $v(t)$.

The two roots of Eq. 8.6 are

$$s_1 = -\frac{1}{2RC} + \sqrt{\left(\frac{1}{2RC}\right)^2 - \frac{1}{LC}}, \tag{8.7}$$

$$s_2 = -\frac{1}{2RC} - \sqrt{\left(\frac{1}{2RC}\right)^2 - \frac{1}{LC}}. \tag{8.8}$$

If either root is substituted into Eq. 8.4, the assumed solution satisfies the given differential equation, that is, Eq. 8.3. Note from Eq. 8.5 that this result holds regardless of the value of A. Therefore, both

$$v = A_1 e^{s_1 t} \text{ and}$$

$$v = A_2 e^{s_2 t}$$

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$$v = v_1 + v_2 = A_1 e^{s_1 t} + A_2 e^{s_2 t}, \tag{8.9}$$

then

$$\frac{dv}{dt} = A_1 s_1 e^{s_1 t} + A_2 s_2 e^{s_2 t}, \tag{8.10}$$

$$\frac{d^2 v}{dt^2} = A_1 s_1^2 e^{s_1 t} + A_2 s_2^2 e^{s_2 t}. \tag{8.11}$$

Substituting Eqs. 8.9–8.11 into Eq. 8.3 gives

$$A_1 e^{s_1 t}\left(s_1^2 + \frac{1}{RC}s_1 + \frac{1}{LC}\right) + A_2 e^{s_2 t}\left(s_2^2 + \frac{1}{RC}s_2 + \frac{1}{LC}\right) = 0. \tag{8.12}$$

But each parenthetical term is zero because by definition s_1 and s_2 are roots of the characteristic equation. Hence the natural response of the parallel RLC circuit shown in Fig. 8.1 is of the form

$$v = A_1 e^{s_1 t} + A_2 e^{s_2 t}. \tag{8.13}$$

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$$v = A_1 e^{s_1 t} + A_2 e^{s_2 t}. \tag{8.13}$$

Equation 8.13 is a repeat of the assumption made in Eq. 8.9. We have shown that v_1 is a solution, v_2 is a solution, and $v_1 + v_2$ is a solution. Therefore, the general solution of Eq. 8.3 has the form given in Eq. 8.13. The roots of the characteristic equation (s_1 and s_2) are determined by the circuit parameters R, L, and C. The initial conditions determine the values of the constants A_1 and A_2. Note that the form of Eq. 8.13 must be modified if the two roots s_1 and s_2 are equal. We discuss this modification when we turn to the critically damped voltage response in Section 8.2.

The behavior of $v(t)$ depends on the values of s_1 and s_2. Therefore the first step in finding the natural response is to determine the roots of the characteristic equation. We return to Eqs. 8.7 and 8.8 and rewrite them using a notation widely used in the literature:

$$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2}, \tag{8.14}$$

$$s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}, \tag{8.15}$$

where

$$\alpha = \frac{1}{2RC}, \tag{8.16}$$

$$\omega_0 = \frac{1}{\sqrt{LC}}. \tag{8.17}$$

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Parameter	Terminology	Value In Natural Response
s_1, s_2	Characteristic roots	$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2}$ $s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}$
α	Neper frequency	$\alpha = \frac{1}{2RC}$
ω_0	Resonant radian frequency	$\omega_0 = \frac{1}{\sqrt{LC}}$

The exponent of e must be dimensionless, so both s_1 and s_2 (and hence α and ω_0) must have the dimension of the reciprocal of time, or frequency. To distinguish among the frequencies s_1, s_2, α, and ω_0, we use the following terminology: s_1 and s_2 are referred to as *complex frequencies*, α is called the *neper frequency*, and ω_0 is the *resonant radian frequency*. The full significance of this terminology unfolds as we move through the remaining chapters of this book. All these frequencies have the dimension of angular frequency per time. For complex frequencies, the neper frequency, and the resonant radian frequency, we specify values using the unit *radians per second* (rad/s). The nature of the roots s_1 and s_2 depends on the values of α and ω_0. There are three possible outcomes. First, if $\omega_0^2 < \alpha^2$, both roots will be real and distinct. For reasons to be discussed later, the voltage response is said to be **overdamped** in this case. Second, if $\omega_0^2 > \alpha^2$, both s_1 and s_2 will be complex and, in addition, will be conjugates of each other. In this situation, the voltage response is said to be **underdamped**. The third possible outcome is that $\omega_0^2 = \alpha^2$. In this case, s_1 and s_2 will be real and equal. Here the voltage response is said to be **critically damped**. As we shall see, damping affects the way the voltage response reaches its final (or steady-state) value. We discuss each case separately in Section 8.2.

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Example 8.1 **Finding the Roots of the Characteristic Equation of a Parallel *RLC* Circuit**

a) Find the roots of the characteristic equation that governs the transient behavior of the voltage shown in Fig. 8.5 if $R = 200\ \Omega$, $L = 50$ mH, and $C = 0.2\ \mu$F.

b) Will the response be overdamped, underdamped, or critically damped?

c) Repeat (a) and (b) for $R = 312.5\ \Omega$.

d) What value of R causes the response to be critically damped?

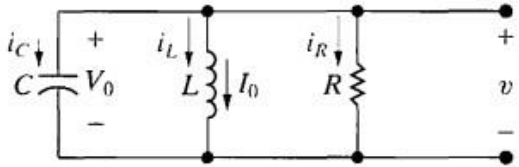


Figure 8.5 ▲ A circuit used to illustrate the natural response of a parallel *RLC* circuit.

Solution

a) For the given values of R, L, and C,

$$\alpha = \frac{1}{2RC} = \frac{10^6}{(400)(0.2)} = 1.25 \times 10^4 \text{ rad/s},$$

$$\omega_0^2 = \frac{1}{LC} = \frac{(10^3)(10^6)}{(50)(0.2)} = 10^8 \text{ rad}^2/\text{s}^2.$$

From Eqs. 8.14 and 8.15,

$$s_1 = -1.25 \times 10^4 + \sqrt{1.5625 \times 10^8 - 10^8}$$

$$= -12{,}500 + 7500 = -5000 \text{ rad/s},$$

$$s_2 = -1.25 \times 10^4 - \sqrt{1.5625 \times 10^8 - 10^8}$$

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b) The voltage response is overdamped because $\omega_0^2 < \alpha^2$.

c) For $R = 312.5\ \Omega$,

$$\alpha = \frac{10^6}{(625)(0.2)} = 8000\ \text{rad/s},$$

$$\alpha^2 = 64 \times 10^6 = 0.64 \times 10^8\ \text{rad}^2/\text{s}^2.$$

As ω_0^2 remains at $10^8\ \text{rad}^2/\text{s}^2$,

$$s_1 = -8000 + j6000\ \text{rad/s},$$

$$s_2 = -8000 - j6000\ \text{rad/s}.$$

(In electrical engineering, the imaginary number $\sqrt{-1}$ is represented by the letter j, because the letter i represents current.)

In this case, the voltage response is underdamped since $\omega_0^2 > \alpha^2$.

d) For critical damping, $\alpha^2 = \omega_0^2$, so

$$\left(\frac{1}{2RC}\right)^2 = \frac{1}{LC} = 10^8,$$

or

$$\frac{1}{2RC} = 10^4,$$

and

$$R = \frac{10^6}{(2 \times 10^4)(0.2)} = 250\ \Omega.$$

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After finding the numerical value of $i_C(0^+)$, we use Eq. 8.21 to find the initial value of dv/dt.

We can summarize the process for finding the overdamped response, $v(t)$, as follows:

1. Find the roots of the characteristic equation, s_1 and s_2, using the values of R, L, and C.
2. Find $v(0^+)$ and $dv(0^+)/dt$ using circuit analysis.
3. Find the values of A_1 and A_2 by solving Eqs. 8.23 and 8.24 simultaneously:

$$v(0^+) = A_1 + A_2, \tag{8.23}$$

$$\frac{dv(0^+)}{dt} = \frac{i_C(0^+)}{C} = s_1A_1 + s_2A_2. \tag{8.24}$$

4. Substitute the values for s_1, s_2, A_1, and A_2 into Eq. 8.18 to determine the expression for $v(t)$ for $t \geq 0$.

The Overdamped Voltage Response

When the roots of the characteristic equation are real and distinct, the voltage response of a parallel RLC circuit is said to be overdamped. The solution for the voltage is of the form

$$v = A_1e^{s_1t} + A_2e^{s_2t}, \tag{8.18}$$

where s_1 and s_2 are the roots of the characteristic equation. The constants A_1 and A_2 are determined by the initial conditions, specifically from the values of $v(0^+)$ and $dv(0^+)/dt$, which in turn are determined from the initial voltage on the capacitor, V_0, and the initial current in the inductor, I_0.

Next, we show how to use the initial voltage on the capacitor and the initial current in the inductor to find A_1 and A_2. First we note from Eq. 8.18 that A_1 and A_2. First we note from Eq. 8.18 that

$$v(0^+) = A_1 + A_2, \tag{8.19}$$

$$\frac{dv(0^+)}{dt} = s_1A_1 + s_2A_2. \tag{8.20}$$

With s_1 and s_2 known, the task of finding A_1 and A_2 reduces to finding $v(0^+)$ and $dv(0^+)/dt$. The value of $v(0^+)$ is the initial voltage on the capacitor V_0. We get the initial value of dv/dt by first finding the current in the capacitor branch at $t = 0^+$. Then,

$$\frac{dv(0^+)}{dt} = \frac{i_C(0^+)}{C}. \tag{8.21}$$

We use Kirchhoff's current law to find the initial current in the capacitor branch. We know that the sum of the three branch currents at $t = 0^+$ must be zero. The current in the resistive branch at $t = 0^+$ is the initial voltage V_0 divided by the resistance, and the current in the inductive branch is I_0. Using the reference system depicted in Fig. 8.5, we obtain

$$i_C(0^+) = \frac{-V_0}{R} - I_0. \tag{8.22}$$

Example 8.2 **Finding the Overdamped Natural Response of a Parallel *RLC* Circuit**

For the circuit in Fig. 8.6, $v(0^+) = 12$ V, and $i_L(0^+) = 30$ mA.

a) Find the initial current in each branch of the circuit.

b) Find the initial value of dv/dt.

c) Find the expression for $v(t)$.

d) Sketch $v(t)$ in the interval $0 \le t \le 250$ ms.

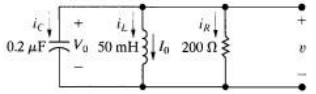


Figure 8.6 ▲ The circuit for Example 8.2.

Solution

a) The inductor prevents an instantaneous change in its current, so the initial value of the inductor current is 30 mA:

$$i_L(0^-) = i_L(0) = i_L(0^+) = 30 \text{ mA}.$$

The capacitor holds the initial voltage across the parallel elements to 12 V. Thus the initial current in the resistive branch, $i_R(0^+)$, is 12/200, or 60 mA. Kirchhoff's current law requires the sum of the currents leaving the top node to equal zero at every instant. Hence

$$\begin{aligned} i_C(0^+) &= -i_L(0^+) - i_R(0^+) \\ &= -90 \text{ mA}. \end{aligned}$$

Note that if we assumed the inductor current and capacitor voltage had reached their dc values at the instant that energy begins to be released, $i_C(0^-) = 0$. In other words, there is an instantaneous change in the capacitor current at $t = 0$.

b) Because $i_C = C(dv/dt)$,

$$\frac{dv(0^+)}{dt} = \frac{-90 \times 10^{-3}}{0.2 \times 10^{-6}} = -450 \text{ kV/s}.$$

c) The roots of the characteristic equation come from the values of R, L, and C. For the values specified and from Eqs. 8.14 and 8.15 along with 8.16 and 8.17,

$$\begin{aligned} s_1 &= -1.25 \times 10^4 + \sqrt{1.5625 \times 10^8 - 10^8} \\ &= -12{,}500 + 7500 = -5000 \text{ rad/s}, \\ s_2 &= -1.25 \times 10^4 - \sqrt{1.5625 \times 10^8 - 10^8} \\ &= -12{,}500 - 7500 = -20{,}000 \text{ rad/s}. \end{aligned}$$

Because the roots are real and distinct, we know that the response is overdamped and hence has the form of Eq. 8.18. We find the co-efficients A_1 and A_2 from Eqs. 8.23 and 8.24. We've already determined s_1, s_2, $v(0^+)$, and $dv(0^+)/dt$, so

$$12 = A_1 + A_2,$$

$$-450 \times 10^3 = -5000A_1 - 20{,}000A_2.$$

We solve two equations for A_1 and A_2 to obtain $A_1 = -14$ V and $A_2 = 26$ V. Substituting these values into Eq. 8.18 yields the overdamped voltage response:

$$v(t) = (-14e^{-5000t} + 26e^{-20{,}000t}) \text{ V}, \quad t \ge 0.$$

As a check on these calculations, we note that the solution yields $v(0) = 12$ V and $dv(0^+)/dt = -450{,}000$ V/s.

d) Figure 8.7 shows a plot of $v(t)$ versus t over the interval $0 \le t \le 250$ ms.

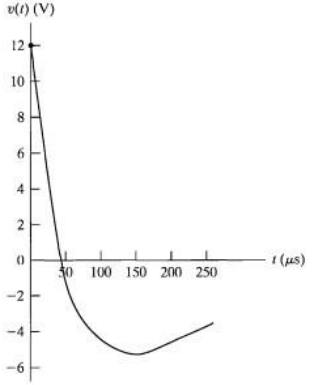


Figure 8.7 ▲ The voltage response for Example 8.2.

The Underdamped Voltage Response

When $\omega_0^2 > \alpha^2$, the roots of the characteristic equation are complex, and the response is underdamped. For convenience, we express the roots s_1 and s_2 as

$$\begin{aligned} s_1 &= -\alpha + \sqrt{-(\omega_0^2 - \alpha^2)} \\ &= -\alpha + j\sqrt{\omega_0^2 - \alpha^2} \\ &= -\alpha + j\omega_d \end{aligned} \tag{8.25}$$

$$s_2 = -\alpha - j\omega_d, \tag{8.26}$$

where

$$\omega_d = \sqrt{\omega_0^2 - \alpha^2}. \tag{8.27}$$

The term ω_d is called the **damped radian frequency**. We explain later the reason for this terminology.

The underdamped voltage response of a parallel RLC circuit is

$$v(t) = B_1 e^{-\alpha t} \cos \omega_d t + B_2 e^{-\alpha t} \sin \omega_d t, \tag{8.28}$$

which follows from Eq. 8.18. In making the transition from Eq. 8.18 to Eq. 8.28, we use the Euler identity:

$$e^{\pm j\theta} = \cos\theta \pm j \sin\theta. \tag{8.29}$$

Thus,

$$\begin{aligned} v(t) &= A_1 e^{(-\alpha + j\omega_d)t} + A_2 e^{-(\alpha + j\omega_d)t} \\ &= A_1 e^{-\alpha t} e^{j\omega_d t} + A_2 e^{-\alpha t} e^{-j\omega_d t} \\ &= e^{-\alpha t}(A_1 \cos \omega_d t + jA_1 \sin \omega_d t + A_2 \cos \omega_d t - jA_2 \sin \omega_d t) \\ &= e^{-\alpha t}[(A_1 + A_2)\cos \omega_d t + j(A_1 - A_2)\sin \omega_d t]. \end{aligned}$$

At this point in the transition from Eq. 8.18 to 8.28, replace the arbitrary constants $A_1 + A_2$ and $j(A_1 - A_2)$ with new arbitrary constants denoted B_1 and B_2 to get

$$\begin{aligned} v &= e^{-\alpha t}(B_1 \cos \omega_d t + B_2 \sin \omega_d t) \\ &= B_1 e^{-\alpha t} \cos \omega_d t + B_2 e^{-\alpha t} \sin \omega_d t. \end{aligned}$$

The constants B_1 and B_2 are real, not complex, because the voltage is a real function. Don't be misled by the fact that $B_2 = j(A_1 - A_2)$. In this underdamped case, A_1 and A_2 are complex conjugates, and thus B_1 and B_2 are real. (See Problems 8.12 and 8.13.) The reason for defining the underdamped response in terms of the coefficients B_1 and B_2 is that it yields a sim-

For the underdamped response, the two simultaneous equations that determine B_1 and B_2 are

$$v(0^+) = V_0 = B_1, \tag{8.30}$$

$$\frac{dv(0^+)}{dt} = \frac{i_c(0^+)}{C} = -\alpha B_1 + \omega_d B_2. \tag{8.31}$$

Let's look at the general nature of the underdamped response. First, the trigonometric functions indicate that this response is oscillatory; that is, the voltage alternates between positive and negative values. The rate at which the voltage oscillates is fixed by ω_d. Second, the amplitude of the oscillation decreases exponentially. The rate at which the amplitude falls off is determined by α. Because α determines how quickly the oscillations subside, it is also referred to as the **damping factor** or **damping coefficient**. That explains why ω_d is called the damped radian frequency. If there is no damping, $\alpha = 0$ and the frequency of oscillation is ω_0. Whenever there is a dissipative element, R, in the circuit, α is not zero and the frequency of oscillation, ω_d, is less than ω_0. Thus when α is not zero, the frequency of oscillation is said to be damped.

The oscillatory behavior is possible because of the two types of energy-storage elements in the circuit: the inductor and the capacitor. (A mechanical analogy of this electric circuit is that of a mass suspended on a spring, where oscillation is possible because energy can be stored in both the spring and the moving mass.) We say more about the characteristics of the underdamped response following Example 8.4, which examines a circuit whose response is underdamped. In summary, note that the overall process for finding the underdamped response is the same as that for the overdamped response, although the response equations and the simultaneous equations used to find the constants are slightly different.

Example 8.4 Finding the Underdamped Natural Response of a Parallel *RLC* Circuit

In the circuit shown in Fig. 8.8, $V_0 = 0$, and $I_0 = -12.25$ mA.

a) Calculate the roots of the characteristic equation.

b) Calculate v and dv/dt at $t = 0^+$.

c) Calculate the voltage response for $t \geq 0$.

d) Plot $v(t)$ versus t for the time interval $0 \leq t \leq 11$ ms.

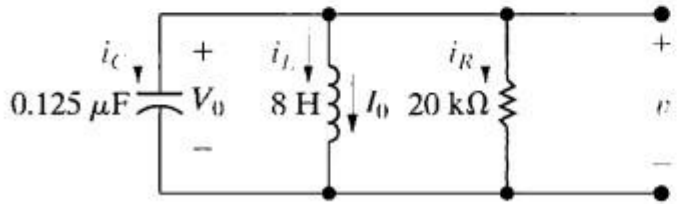


Figure 8.8 ▲ The circuit for Example 8.4.

Solution

a) Because

$$\alpha = \frac{1}{2RC} = \frac{10^6}{2(20)10^3(0.125)} = 200 \text{ rad/s},$$

$$\omega_0 = \frac{1}{\sqrt{LC}} = \sqrt{\frac{10^6}{(8)(0.125)}} = 10^3 \text{ rad/s},$$

we have

$$\omega_0^2 > \alpha^2.$$

Therefore, the response is underdamped. Now,

$$\omega_d = \sqrt{\omega_0^2 - \alpha^2} = \sqrt{10^6 - 4 \times 10^4} = 100\sqrt{96}$$

$$= 979.80 \text{ rad/s},$$

$$s_1 = -\alpha + j\omega_d = -200 + j979.80 \text{ rad/s},$$

$$s_2 = -\alpha - j\omega_d = -200 - j979.80 \text{ rad/s}.$$

For the underdamped case, we do not ordinarily solve for s_1 and s_2 because we do not use them explicitly. However, this example emphasizes why s_1 and s_2 are known as complex frequencies.

b) Because v is the voltage across the terminals of a capacitor, we have

$$v(0) = v(0^+) = V_0 = 0.$$

Because $v(0^+) = 0$, the current in the resistive branch is zero at $t = 0^+$. Hence the current in the capacitor at $t = 0^+$ is the negative of the inductor current:

$$i_C(0^+) = -(-12.25) = 12.25 \text{ mA}.$$

Therefore the initial value of the derivative is

$$\frac{dv(0^+)}{dt} = \frac{(12.25)(10^{-3})}{(0.125)(10^{-6})} = 98{,}000 \text{ V/s}.$$

c) From Eqs. 8.30 and 8.31, $B_1 = 0$ and

$$B_2 = \frac{98{,}000}{\omega_d} \approx 100 \text{ V}.$$

Substituting the numerical values of α, ω_d, B_1, and B_2 into the expression for $v(t)$ gives

$$v(t) = 100e^{-200t} \sin 979.80t \text{ V}, \quad t \geq 0.$$

d) Figure 8.9 shows the plot of $v(t)$ versus t for the first 11 ms after the stored energy is released. It clearly indicates the damped oscillatory nature of the underdamped response. The voltage $v(t)$ approaches its final value, alternating between values that are greater than and less than the final value. Furthermore, these swings about the final value decrease exponentially with time.

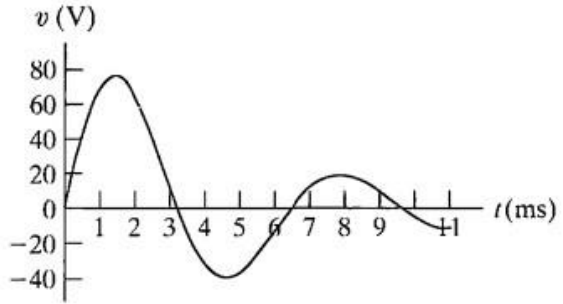


Figure 8.9 ▲ The voltage response for Example 8.4.

The Critically Damped Voltage Response

The second-order circuit in Fig. 8.8 is critically damped when $\omega_0^2 = \alpha^2$, or $\omega_0 = \alpha$. When a circuit is critically damped, the response is on the verge of oscillating. In addition, the two roots of the characteristic equation are real and equal; that is,

$$s_1 = s_2 = -\alpha = -\frac{1}{2RC}. \tag{8.32}$$

When this occurs, the solution for the voltage no longer takes the form of Eq. 8.18. This equation breaks down because if $s_1 = s_2 = -\alpha$, it predicts that

$$v = (A_1 + A_2)e^{-\alpha t} = A_0 e^{-\alpha t}, \tag{8.33}$$

where A_0 is an arbitrary constant. Equation 8.33 cannot satisfy two independent initial conditions (V_0, I_0) with only one arbitrary constant, A_0. Recall that the circuit parameters R and C fix α.

We can trace this dilemma back to the assumption that the solution takes the form of Eq. 8.18. When the roots of the characteristic equation are equal, the solution for the differential equation takes a different form, namely

$$v(t) = D_1 t e^{-\alpha t} + D_2 e^{-\alpha t}. \tag{8.34}$$

From Eq. 8.34, the two simultaneous equations needed to determine D_1 and D_2 are

$$v(0^+) = V_0 = D_2, \tag{8.35}$$

$$\frac{dv(0^+)}{dt} = \frac{i_C(0^+)}{C} = D_1 - \alpha D_2. \tag{8.36}$$

Example 8.5 Finding the Critically Damped Natural Response of a Parallel *RLC* Circuit

a) For the circuit in Example 8.4 (Fig. 8.8), find the value of R that results in a critically damped voltage response.

b) Calculate $v(t)$ for $t \geq 0$.

c) Plot $v(t)$ versus t for $0 \leq t \leq 7$ ms.

Solution

a) From Example 8.4, we know that $\omega_0^2 = 10^6$. Therefore for critical damping,

$$\alpha = 10^3 = \frac{1}{2RC},$$

or

$$R = \frac{10^6}{(2000)(0.125)} = 4000\ \Omega.$$

b) From the solution of Example 8.4, we know that $v(0^+) = 0$ and $dv(0^+)/dt = 98{,}000$ V/s. From Eqs. 8.35 and 8.36, $D_2 = 0$ and $D_1 = 98{,}000$ V/s. Substituting these values for a, D_1, and D_2 into Eq. 8.34 gives

$$v(t) = 98{,}000te^{-1000t}\ \text{V}, \quad t \geq 0.$$

c) Figure 8.10 shows a plot of $v(t)$ versus t in the interval $0 \leq t \leq 7$ ms.

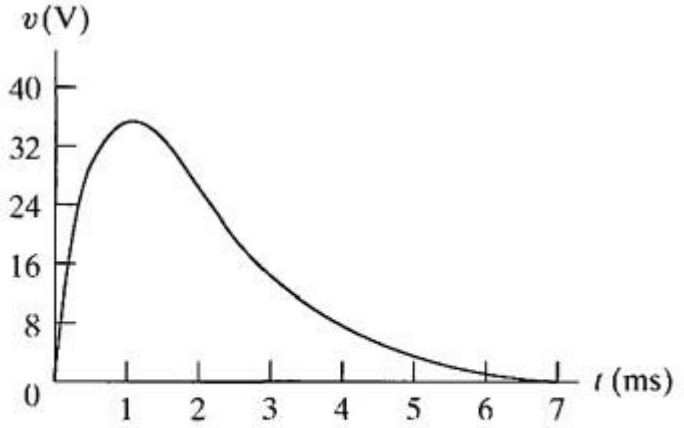


Figure 8.10 ▲ The voltage response for Example 8.5.

A Summary of the Results

We conclude our discussion of the parallel *RLC* circuit's natural response with a brief summary of the results. The first step in finding the natural response is to calculate the roots of the characteristic equation. You then know immediately whether the response is overdamped, underdamped, or critically damped.

If the roots are real and distinct ($\omega_0^2 < \alpha^2$), the response is overdamped and the voltage is

$$v(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t},$$

where

$$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2},$$

$$s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2},$$

$$\alpha = \frac{1}{2RC},$$

$$\omega_0^2 = \frac{1}{LC}.$$

The values of A_1 and A_2 are determined by solving the following simultaneous equations:

$$v(0^+) = A_1 + A_2,$$

$$\frac{dv(0^+)}{dt} = \frac{i_C(0^+)}{C} = s_1 A_1 + s_2 A_2.$$

If the roots are complex $\omega_0^2 > \alpha^2$ the response is underdamped and the voltage is

$$v(t) = B_1 e^{-\alpha t} \cos \omega_d t + B_2 e^{-\alpha t} \sin \omega_d t,$$

where

$$\omega_d = \sqrt{\omega_0^2 - \alpha^2}.$$

The values of B_1 and B_2 are found by solving the following simultaneous equations:

$$v(0^+) = V_0 = B_1,$$

$$\frac{dv(0^+)}{dt} = \frac{i_C(0^+)}{C} = -\alpha B_1 + \omega_d B_2.$$

If the roots of the characteristic equation are real and equal ($\omega_0^2 = \alpha^2$), the voltage response is

$$v(t) = D_1 t e^{-\alpha t} + D_2 e^{-\alpha t},$$

where α is as in the other solution forms. To determine values for the constants D_1 and D_2, solve the following simultaneous equations:

$$v(0^+) = V_0 = D_2,$$

$$\frac{dv(0^+)}{dt} = \frac{i_C(0^+)}{C} = D_1 - \alpha D_2.$$

8.3 The Step Response of a Parallel *RLC* Circuit

Finding the step response of a parallel *RLC* circuit involves finding the voltage across the parallel branches or the current in the individual branches as a result of the sudden application of a dc current source. There may or may not be energy stored in the circuit when the current source is applied. The task is represented by the circuit shown in Fig. 8.11. To develop a general approach to finding the step response of a second-order circuit, we focus on finding the current in the inductive branch (i_L). This current is of particular interest because it does not approach zero as t increases. Rather, after the switch has been open for a long time, the inductor current equals the dc source current I. Because we want to focus on the technique for finding the step response, we assume that the initial energy stored in the circuit is zero. This assumption simplifies the calculations and doesn't alter the basic process involved. In Example 8.10 we will see how the presence of initially stored energy enters into the general procedure.

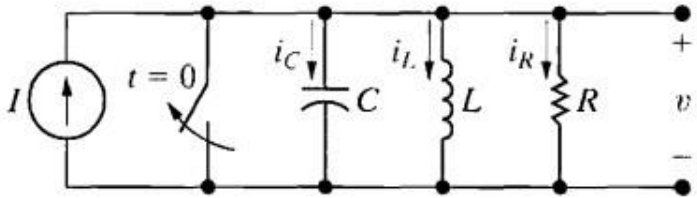


Figure 8.11 ▲ A circuit used to describe the step response of a parallel *RLC* circuit.

To find the inductor current i_L, we must solve a second-order differential equation equated to the forcing function I, which we derive as follows. From Kirchhoff's current law, we have

$$i_L + i_R + i_C = I,$$

or

$$i_L + \frac{v}{R} + C\frac{dv}{dt} = I. \tag{8.37}$$

Because

$$v = L\frac{di_L}{dt}, \tag{8.38}$$

we get

$$\frac{dv}{dt} = L\frac{d^2i_L}{dt^2}. \tag{8.39}$$

Substituting Eqs. 8.38 and 8.39 into Eq. 8.37 gives

$$i_L + \frac{L}{R}\frac{di_L}{dt} + LC\frac{d^2i_L}{dt^2} = I. \tag{8.40}$$

For convenience, we divide through by LC and rearrange terms:

$$\frac{d^2i_L}{dt^2} + \frac{1}{RC}\frac{di_L}{dt} + \frac{i_L}{LC} = \frac{I}{LC}. \tag{8.41}$$

The Direct Approach

It is much easier to find the primed constants directly in terms of the initial values of the response function. For the circuit being discussed, we would find the primed constants from $i_L(0)$ and $di_L(0)/dt$.

The solution for a second-order differential equation with a constant forcing function equals the forced response plus a response function identical in form to the natural response. Thus we can always write the solution for the step response in the form

$$i = I_f + \left\{\begin{array}{c}\text{function of the same form} \\ \text{as the natural response}\end{array}\right\}, \tag{8.50}$$

or

$$v = V_f + \left\{\begin{array}{c}\text{function of the same form} \\ \text{as the natural response}\end{array}\right\}, \tag{8.51}$$

where I_f and V_f represent the final value of the response function. The final value may be zero, as was, for example, the case with the voltage v in the circuit in Fig. 8.8.

Examples 8.6–8.10 illustrate the technique of finding the step response of a parallel RLC circuit using the direct approach.

Example 8.6 **Finding the Overdamped Step Response of a Parallel *RLC* Circuit**

The initial energy stored in the circuit in Fig. 8.12 is zero. At $t = 0$, a dc current source of 24 mA is applied to the circuit. The value of the resistor is 400 Ω.

a) What is the initial value of i_L?

b) What is the initial value of di_L/dt?

c) What are the roots of the characteristic equation?

d) What is the numerical expression for $i_L(t)$ when $t \geq 0$?

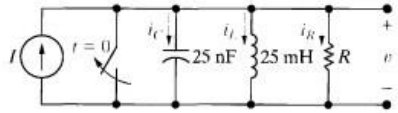


Figure 8.12 ▲ The circuit for Example 8.6.

Solution

a) No energy is stored in the circuit prior to the application of the dc current source, so the initial current in the inductor is zero. The inductor prohibits an instantaneous change in inductor current; therefore $i_L(0) = 0$ immediately after the switch has been opened.

b) The initial voltage on the capacitor is zero before the switch has been opened; therefore it will be zero immediately after. Now, because $v = L di_L/dt$,

$$\frac{di_L}{dt}(0^+) = 0.$$

c) From the circuit elements, we obtain

$$\omega_0^2 = \frac{1}{LC} = \frac{10^{12}}{(25)(25)} = 16 \times 10^8,$$

$$\alpha = \frac{1}{2RC} = \frac{10^9}{(2)(400)(25)} = 5 \times 10^4 \text{ rad/s},$$

or

$$\alpha^2 = 25 \times 10^8.$$

Because $\omega_0^2 < \alpha^2$, the roots of the characteristic equation are real and distinct. Thus

$$s_1 = -5 \times 10^4 + 3 \times 10^4 = -20{,}000 \text{ rad/s},$$

$$s_2 = -5 \times 10^4 - 3 \times 10^4 = -80{,}000 \text{ rad/s}.$$

d) Because the roots of the characteristic equation are real and distinct, the inductor current response will be overdamped. Thus $i_L(t)$ takes the form of Eq. 8.47, namely,

$$i_L = I_f + A_1' e^{s_1 t} + A_2' e^{s_2 t}.$$

▶ **Inductor current in overdamped parallel *RLC* circuit step response**

Hence, from this solution, the two simultaneous equations that determine A_1' and A_2' are

$$i_L(0) = I_f + A_1' + A_2' = 0,$$

$$\frac{di_L}{dt}(0) = s_1 A_1' + s_2 A_2' = 0.$$

Solving for A_1' and A_2' gives

$$A_1' = -32 \text{ mA} \quad \text{and} \quad A_2' = 8 \text{ mA}.$$

The numerical solution for $i_L(t)$ is

$$i_L(t) = (24 - 32e^{-20{,}000t} + 8e^{-80{,}000t}) \text{ mA}, \quad t \geq 0.$$

Example 8.7 **Finding the Underdamped Step Response of a Parallel *RLC* Circuit**

The resistor in the circuit in Example 8.6 (Fig. 8.12) is increased to 625 Ω. Find $i_L(t)$ for $t \geq 0$.

Solution

Because L and C remain fixed, ω_0^2 has the same value as in Example 8.6; that is, $\omega_0^2 = 16 \times 10^8$. Increasing R to 625 Ω decreases α to 3.2×10^4 rad/s. With $\omega_0^2 > \alpha^2$, the roots of the characteristic equation are complex. Hence

$$s_1 = -3.2 \times 10^4 + j2.4 \times 10^4 \text{ rad/s},$$

$$s_2 = -3.2 \times 10^4 - j2.4 \times 10^4 \text{ rad/s}.$$

The current response is now underdamped and given by Eq. 8.48:

$$i_L(t) = I_f + B_1' e^{-\alpha t} \cos \omega_d t + B_2' e^{-\alpha t} \sin \omega_d t.$$

▶ **Inductor current in underdamped parallel *RLC* circuit step response**

Here, α is 32,000 rad/s, ω_d is 24,000 rad/s, and I_f is 24 mA.

As in Example 8.6, B_1' and B_2' are determined from the initial conditions. Thus the two simultaneous equations are

$$i_L(0) = I_f + B_1' = 0,$$

$$\frac{di_L}{dt}(0) = \omega_d B_2' - \alpha B_1' = 0.$$

Then,

$$B_1' = -24 \text{ mA}$$

and

$$B_2' = -32 \text{ mA}.$$

The numerical solution for $i_L(t)$ is

$$i_L(t) = (24 - 24e^{-32{,}000t} \cos 24{,}000t - 32e^{-32{,}000t} \sin 24{,}000t) \text{ mA}, \quad t \geq 0.$$

Example 8.8 **Finding the Critically Damped Step Response of a Parallel *RLC* Circuit**

The resistor in the circuit in Example 8.6 (Fig. 8.12) is set at 500 Ω. Find i_L for $t \geq 0$.

Solution

We know that ω_0^2 remains at 16×10^8. With R set at 500 Ω, α becomes $4 \times 10^4\ s^{-1}$, which corresponds to critical damping. Therefore the solution for $i_L(t)$ takes the form of Eq. 8.49:

$$i_L(t) = I_f + D_1' te^{-\alpha t} + D_2' e^{-\alpha t}.$$

▶ **Inductor current in critically damped parallel *RLC* circuit step response**

Again, D_1' and D_2' are computed from initial conditions, or

$$i_L(0) = I_f + D_2' = 0,$$

$$\frac{di_L}{dt}(0) = D_1' - \alpha D_2' = 0.$$

Thus

$$D_1' = -960{,}000\ \text{mA/s} \quad \text{and} \quad D_2' = -24\ \text{mA}.$$

The numerical expression for $i_L(t)$ is

$$i_L(t) = (24 - 960{,}000te^{-40{,}000t} - 24e^{-40{,}000t})\ \text{mA},\ t \geq 0.$$

Example 8.9 **Comparing the Three-Step Response Forms**

a) Plot on a single graph, over a range from 0 to 220 μs, the overdamped, underdamped, and critically damped responses derived in Examples 8.6–8.8.

b) Use the plots of (a) to find the time required for i_L to reach 90% of its final value.

c) On the basis of the results obtained in (b), which response would you specify in a design that puts a premium on reaching 90% of the final value of the output in the shortest time?

d) Which response would you specify in a design that must ensure that the final value of the current is never exceeded?

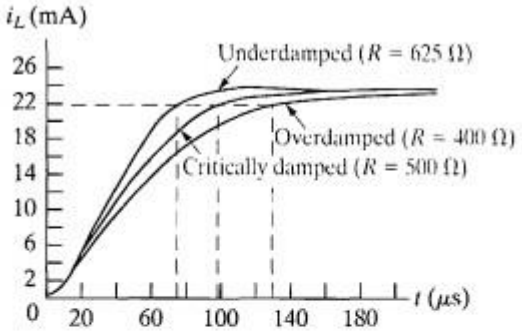


Figure 8.13 ▲ The current plots for Example 8.9.

Solution

a) See Fig. 8.13.

b) The final value of i_L is 24 mA, so we can read the times off the plots corresponding to i_L = 21.6 mA. Thus t_{od} = 130 μs, t_{cd} = 97 μs, and t_{ud} = 74 μs.

c) The underdamped response reaches 90% of the final value in the fastest time, so it is the desired response type when speed is the most important design specification.

d) From the plot, you can see that the underdamped response overshoots the final value of current, whereas neither the critically damped nor the overdamped response produces currents in excess of 24 mA. Although specifying either of the latter two responses would meet the design specification, it is best to use the overdamped response. It would be impractical to require a design to achieve the exact component values that ensure a critically damped response.

8.4 The Natural and Step Response of a Series *RLC* Circuit

The procedures for finding the natural or step responses of a series *RLC* circuit are the same as those used to find the natural or step responses of a parallel *RLC* circuit, because both circuits are described by differential equations that have the same form. We begin by summing the voltages around the closed path in the circuit shown in Fig. 8.14. Thus

$$Ri + L\frac{di}{dt} + \frac{1}{C}\int_0^t i\,d\tau + V_0 = 0. \tag{8.52}$$

We now differentiate Eq. 8.52 once with respect to t to get

$$R\frac{di}{dt} + L\frac{d^2i}{dt^2} + \frac{i}{C} = 0, \tag{8.53}$$

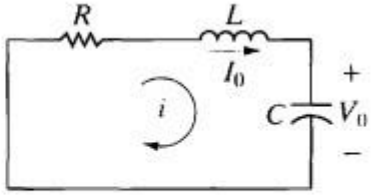


Figure 8.14 ▲ A circuit used to illustrate the natural response of a series *RLC* circuit.

which we can rearrange as

$$\frac{d^2i}{dt^2} + \frac{R}{L}\frac{di}{dt} + \frac{i}{LC} = 0. \tag{8.54}$$

Comparing Eq. 8.54 with Eq. 8.3 reveals that they have the same form. Therefore, to find the solution of Eq. 8.54, we follow the same process that led us to the solution of Eq. 8.3.

From Eq. 8.54, the characteristic equation for the series *RLC* circuit is

$$s^2 + \frac{R}{L}s + \frac{1}{LC} = 0. \tag{8.55}$$

The roots of the characteristic equation are

$$s_{1,2} = -\frac{R}{2L} \pm \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}, \tag{8.56}$$

or

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}. \tag{8.57}$$

The neper frequency (α) for the series *RLC* circuit is

$$\alpha = \frac{R}{2L} \text{ rad/s}, \tag{8.58}$$

and the expression for the resonant radian frequency is

$$\omega_0 = \frac{1}{\sqrt{LC}} \text{ rad/s}. \tag{8.59}$$

Note that the equation for neper frequency of the series *RLC* circuit differs from that of the parallel *RLC* circuit, but the equations for resonant and damped radian frequencies are the same.

The current response will be overdamped, underdamped, or critically damped according to whether $\omega_0^2 < \alpha^2$, $\omega_0^2 > \alpha^2$, or $\omega_0^2 = \alpha^2$, respectively. Thus the three possible solutions for the current are as follows:

$$i(t) = A_1e^{s_1t} + A_2e^{s_2t} \text{ (overdamped)}, \tag{8.60}$$

$$i(t) = B_1e^{-\alpha t}\cos\omega_d t + B_2e^{-\alpha t}\sin\omega_d t \text{ (underdamped)}, \tag{8.61}$$

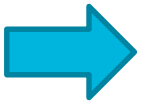
$$i(t) = D_1te^{-\alpha t} + D_2e^{-\alpha t} \text{ (critically damped)}. \tag{8.62}$$

When you have obtained the natural current response, you can find the natural voltage response across any circuit element.

To verify that the procedure for finding the step response of a series *RLC* circuit is the same as that for a parallel *RLC* circuit, we show that the differential equation that describes the capacitor voltage in Fig. 8.15 has the same form as the differential equation that describes the inductor current in Fig. 8.11. For convenience, we assume that zero energy is stored in the circuit at the instant the switch is closed.

Applying Kirchhoff's voltage law to the circuit shown in Fig. 8.15 gives

$$V = Ri + L\frac{di}{dt} + v_C. \tag{8.63}$$



The current (i) is related to the capacitor voltage (v_C) by the expression

$$i = C\frac{dv_C}{dt}, \tag{8.64}$$

from which

$$\frac{di}{dt} = C\frac{d^2v_C}{dt^2}. \tag{8.65}$$

Substitute Eqs. 8.64 and 8.65 into Eq. 8.63 and write the resulting expression as

$$\frac{d^2v_C}{dt^2} + \frac{R}{L}\frac{dv_C}{dt} + \frac{v_C}{LC} = \frac{V}{LC}. \tag{8.66}$$

Equation 8.66 has the same form as Eq. 8.41; therefore the procedure for finding v_C parallels that for finding i_L. The three possible solutions for v_C are as follows:

$$v_C = V_f + A_1'e^{s_1t} + A_2'e^{s_2t} \text{ (overdamped)}, \tag{8.67}$$

$$v_C = V_f + B_1'e^{-\alpha t}\cos\omega_d t + B_2'e^{-\alpha t}\sin\omega_d t \text{ (underdamped)}, \tag{8.68}$$

$$v_C = V_f + D_1'te^{-\alpha t} + D_2'e^{-\alpha t} \text{ (critically damped)}, \tag{8.69}$$

where V_f is the final value of v_C. Hence, from the circuit shown in Fig. 8.15, the final value of v_C is the dc source voltage V.

Example 8.11 and 8.12 illustrate the mechanics of finding the natural and step responses of a series *RLC* circuit.

Example 8.11 Finding the Underdamped Natural Response of a Series *RLC* Circuit

The 0.1 μF capacitor in the circuit shown in Fig. 8.16 is charged to 100 V. At $t = 0$ the capacitor is discharged through a series combination of a 100 mH inductor and a 560 Ω resistor.

a) Find $i(t)$ for $t \geq 0$.

b) Find $v_C(t)$ for $t \geq 0$.

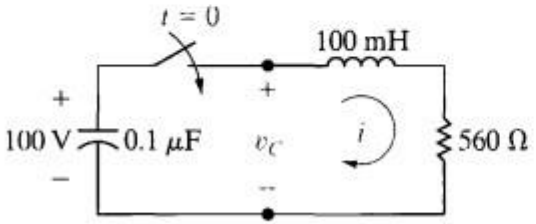


Figure 8.16 ▲ The circuit for Example 8.11.

Solution

a) The first step to finding $i(t)$ is to calculate the roots of the characteristic equation. For the given element values,

$$\omega_0^2 = \frac{1}{LC}$$

$$= \frac{(10^3)(10^6)}{(100)(0.1)} = 10^8,$$

$$\alpha = \frac{R}{2L}$$

$$= \frac{560}{2(100)} \times 10^3$$

$$= 2800 \text{ rad/s}.$$

Next, we compare ω_0^2 to α^2 and note that $\omega_0^2 > \alpha^2$, because

$$\alpha^2 = 7.84 \times 10^6$$

$$= 0.0784 \times 10^8.$$

At this point, we know that the response is underdamped and that the solution for $i(t)$ is of the form

$$i(t) = B_1 e^{-\alpha t} \cos \omega_d t + B_2 e^{-\alpha t} \sin \omega_d t,$$

where $\alpha = 2800$ rad/s and $\omega_d = 9600$ rad/s. The numerical values of B_1 and B_2 come from the initial conditions. The inductor current is zero before the switch has been closed, and hence it is zero immediately after. Therefore

$$i(0) = 0 = B_1.$$

To find B_2, we evaluate $di(0^+)/dt$. From the circuit, we note that, because $i(0) = 0$ immediately after the switch has been closed, there will be no voltage drop across the resistor. Thus the initial voltage on the capacitor appears across the terminals of the inductor, which leads to the expression,

$$L\frac{di(0^+)}{dt} = V_0,$$

or

$$\frac{di(0^+)}{dt} = \frac{V_0}{L} = \frac{100}{100} \times 10^3$$

$$= 1000 \text{ A/s}.$$

Because $B_1 = 0$,

$$\frac{di}{dt} = 400B_2 e^{-2800t}(24 \cos 9600t - 7 \sin 9600t).$$

Thus

$$\frac{di(0^+)}{dt} = 9600B_2,$$

$$B_2 = \frac{1000}{9600} \approx 0.1042 \text{ A}.$$

The solution for $i(t)$ is

$$i(t) = 0.1042e^{-2800t} \sin 9600t \text{ A}, \quad t \geq 0.$$

b) To find $v_C(t)$, we can use either of the following relationships:

$$v_C = -\frac{1}{C}\int_0^t i\,d\tau + 100 \text{ or}$$

$$v_C = iR + L\frac{di}{dt}.$$

Whichever expression is used (the second is recommended), the result is

$$v_C(t) = (100 \cos 9600t + 29.17 \sin 9600t)e^{-2800t} \text{ V}, \quad t \geq 0.$$

Example 8.12 Finding the Underdamped Step Response of a Series *RLC* Circuit

No energy is stored in the 100 mH inductor or the 0.4 μF capacitor when the switch in the circuit shown in Fig. 8.17 is closed. Find $v_C(t)$ for $t \geq 0$.

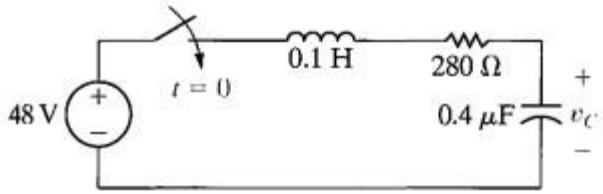


Figure 8.17 ▲ The circuit for Example 8.12.

Solution

The roots of the characteristic equation are

$$s_1 = -\frac{280}{0.2} + \sqrt{\left(\frac{280}{0.2}\right)^2 - \frac{10^6}{(0.1)(0.4)}}$$

$$= (-1400 + j4800) \text{ rad/s},$$

$$s_2 = (-1400 - j4800) \text{ rad/s}.$$

The roots are complex, so the voltage response is underdamped. Thus

$$v_C(t) = 48 + B_1' e^{-1400t} \cos 4800t + B_2' e^{-1400t} \sin 4800t, \quad t \geq 0.$$

No energy is stored in the circuit initially, so both $v_C(0)$ and $dv_C(0^+)/dt$ are zero. Then,

$$v_C(0) = 0 = 48 + B_1',$$

$$\frac{dv_C(0^+)}{dt} = 0 = 4800B_2' - 1400B_1'.$$

Solving for B_1' and B_2' yields

$$B_1' = -48 \text{ V},$$

$$B_2' = -14 \text{ V}.$$

Therefore, the solution for $v_C(t)$ is

$$v_C(t) = (48 - 48e^{-1400t} \cos 4800t - 14e^{-1400t} \sin 4800t) \text{ V}, \quad t \geq 0.$$

TABLE 8.2 The Response of a Second-Order Circuit is Overdamped, Underdamped, or Critically Damped

The Circuit is	When	Qualitative Nature of the Response
Overdamped	$\alpha^2 > \omega_0^2$	The voltage or current approaches its final value without oscillation
Underdamped	$\alpha^2 < \omega_0^2$	The voltage or current oscillates about its final value
Critically damped	$\alpha^2 = \omega_0^2$	The voltage or current is on the verge of oscillating about its final value

TABLE 8.3 In Determining the Natural Response of a Second-Order Circuit, We First Determine Whether it is Over-, Under-, or Critically Damped, and Then We Solve the Appropriate Equations

Damping	Natural Response Equations	Coefficient Equations
Overdamped	$x(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$	$x(0) = A_1 + A_2$; $dx/dt(0) = A_1 s_1 + A_2 s_2$
Underdamped	$x(t) = (B_1 \cos \omega_d t + B_2 \sin \omega_d t) e^{-\alpha t}$	$x(0) = B_1$; $dx/dt(0) = -\alpha B_1 + \omega_d B_2$, where $\omega_d = \sqrt{\omega_0^2 - \alpha^2}$
Critically damped	$x(t) = (D_1 t + D_2) e^{-\alpha t}$	$x(0) = D_2$, $dx/dt(0) = D_1 - \alpha D_2$

TABLE 8.4 In Determining the Step Response of a Second-Order Circuit, We Apply the Appropriate Equations Depending on the Damping

Damping	Step Response Equations[a]	Coefficient Equations
Overdamped	$x(t) = X_f + A_1' e^{s_1 t} + A_2' e^{s_2 t}$	$x(0) = X_f + A_1' + A_2'$; $dx/dt(0) = A_1' s_1 + A_2' s_2$
Underdamped	$x(t) = X_f + (B_1' \cos \omega_d t + B_2' \sin \omega_d t)e^{-\alpha t}$	$x(0) = X_f + B_1'$; $dx/dt(0) = -\alpha B_1' + \omega_d B_2'$
Critically damped	$x(t) = X_f + D_1' te^{-\alpha t} + D_2' e^{-\alpha t}$	$x(0) = X_f + D_2'$; $dx/dt(0) = D_1' - \alpha D_2'$

[a] where X_f is the final value of $x(t)$.

End of the Lesson (09.01.24)

Any questions?

New supplementary videos are added to the playlist on Youtube:

Supplementary videos: Enter Youtube-> Search for ‘Alparslan Zehir’ -> Go to Playlists -> Watch ‘Electrical Circuits’ playlist